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Single- and Multilayers of Alkali Metal Atoms inside Graphene/MoS_2 Heterostructures: A Systematic First-Principles Study

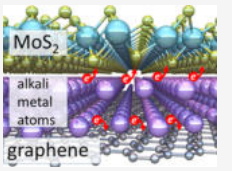
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Cite This: *J. Phys. Chem. C* 2022, 126, 15558−15564

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ABSTRACT: Stacking various 2D materials in van der Waals heterostructures is a novel approach to design new systems, which can host alkali metal (AM) atoms to tune their electronic properties or store energy. Using state-of-the-art first-principles calculations, we systematically study the intercalation of the most widespread AMs (Li, Na, and K) into a graphene/MoS_2 heterostructure. Contrary to the previous work on the intercalation of AMs into various heterostructures based on 2D materials, we consider not only single-, but also multi-layer configurations of AM atoms. We assess the intercalation energetics for various concentrations of AM atoms, calculate charge transfer from AM atoms to the host system, and show that although intercalation of AMs as a single layer is energetically preferable, multi-layer configurations can exist at high concentrations of AM atoms. We further demonstrate that the transition of the MoS_2 layer from the H to T′ phase is possible upon Li intercalation, but not for Na or K. Our findings should help to better understand the behavior of heterostructures upon AM atom intercalation and may stimulate further experiments aimed at the tailoring of heterostructure properties and increasing the capacity of anode materials in AM ion batteries.



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INTRODUCTION

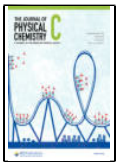
The intercalation of alkali metal (AM) atoms into layered bulk and low-dimensional materials has recently received an enormous amount of attention in the context of energy storage[1−3] and tuning the opto-electronic properties of the materials.[4−6] As for the former, increasing the capacity of metal-ion batteries is believed to be crucial for further development of portable electronics and electrical vehicles. One of the promising directions of improving the characteristics of metal-ion batteries, such as the capacity and the critical number of cycles, is the use of composites as anodes.[7−10] Among these systems, van der Waals (vdW) heterostructures consisting of stacked layers of graphene and transition metal dichalcogenides (TMDs), such as MoS_2, have attracted particular attention[11−14] as perspective metal anode materials. MoS_2 has low electronic conductivity and undergoes a strong volume change upon intercalation/deintercalation, but intercalation of AMs into these systems is energetically more favorable as compared to graphitic structures,[15] so that a combination of MoS_2 with conductive carbon matrices is expected to be a promising direction. In particular, MoS_2/graphene heterostructures were synthesized[11−14] and demonstrated to be promising systems as anode materials in lithium-ion batteries exhibiting extraordinary capacity (up to 1300 mA h g^{-1}) and excellent rate capability and cycling stability.

With regard to tailoring characteristics of heterostructures, Li intercalation into electrochemical devices based on stacked hexagonal boron nitride, TMDs, and graphene layers has been demonstrated to be an effective approach to control the electronic and optical properties of such systems.[4] As for MoS_2/graphene heterostructures, intercalated Li atoms have been found to increase the electronic conductivity and electrochemical performance of these systems.[11−13]

The AM intercalation into heterostructures has also been theoretically investigated. Specifically, studies focused on the intercalation of AMs into MoS_2/graphene and other heterostructures have addressed the stability, charge transfer, and electronic structure of these systems.[15−18] In particular, it was shown that intercalation of Li between Gr/MoS_2 layers is energetically more favorable over adsorption on either side[15] of the structure. It was also shown[17] that an increase in the concentration of intercalated Li atoms in Gr/MoS_2 composite leads to band gap opening in graphene. Larson et al. studied the intercalation of lithium atoms into Gr/MoS_2 heterostructures with different MoS_2 phases (H and T′) and determined that the morphology of MoS_2 does not significantly affect the stability, Li diffusion barrier, and charge transfer.[18]

Na intercalation into hybrid structures has been studied as well,[19−22] as Na-ion batteries, if developed, are expected to be economically and environmentally much more attractive than Li-ion batteries. Massaro et al.[23] carried out first-principles calculations aimed at the investigation of sodium intercalation between graphene and MoS_2 layers, assessed the electronic properties and possible diffusion paths of Na atoms along the

Received: June 1, 2022
Revised: August 25, 2022
Published: September 7, 2022



ACS Publications

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interfaces, and estimated the corresponding energy barriers. In addition to Na, K has also been actively considered as an alternative to Li. Recently, the possibility of creating K-ion batteries has been intensely studied.[24,25] Zhang et al. investigated K and Li intercalation into N-doped graphene/MoS_2 (2H and 1T) and showed that doping plays a vital role in improving the structural stability of Gr/MoS_2 structures.[26]

However, many aspects of AM intercalation in heterostructures remain not fully understood. In particular, the energetics and possible phase segregation of AM atoms as a function of their concentration have not been studied for Na and K. Many investigations were focused on bulk structures, not Gr/MoS_2 bilayers, whereas recent experiments demonstrated that formation of multi-layer Li structures is possible in bilayer graphene.[27] Calculations also predict that this should be possible for some AMs (at least Li and K) intercalated into MoS_2 bilayer.[28]

In this work, we systematically study the intercalation of Li, Na, and K into a Gr/MoS_2 bilayer heterostructure using first-principles calculations. We consider both single- and multi-layer AM configurations and investigate the energetics related to the stability of the system for various concentrations of AMs. We further address the charge transfer and possible transformation of MoS_2 from the H to T′ phase.

SIMULATION METHODOLOGY

All our calculations were carried out within the framework of the density functional theory (DFT) as implemented in the VASP software package.[29,30] An energy cutoff of 600 eV was used for the supercell calculations. The Brillouin zones of the primitive cells of the 2D materials and bulk AMs were sampled using 12 × 12 × 1 and 12 × 12 × 12 Monkhorst−Pack grid points.[31] A Γ-centered k-point 5 × 5 × 1 sampling was used in the supercell calculations. The maximum force on each atom was set to be less than 0.01 eV for the geometry optimization. All calculations were performed on a 5 × 5 graphene 4 × 4 MoS_2 periodic supercell structure, consisting of 98 atoms. The concentration of Li, Na, and K atoms changed either from 1 to 16 (for Na and K) or to 19 (for Li). Dipole corrections were not used in the calculations, but test runs with account for the corrections gave the values of the total energy being different by less than 0.01 eV. The atomic structures and the charge densities were visualized using the VESTA package.[32] At least 20 Å of vacuum was used along the direction perpendicular to the layers (z-axis) to prevent spurious image interaction. To build the Gr/MoS_2 structures with multi-layered AMs, we used the Virtual NanoLab software.[33] The typical atomic configurations for single- and multi-layer AM structures are presented in Figure 1.

To estimate the energetics directly related to the stability of different configurations, we calculated the intercalation energy E_i per AM atom, corresponding to the energy change when n_{AM} AM atoms are moved from the bulk AM crystal into the heterostructure. It is defined as

$$E_i = [E(n_{AM}) - E(0)]/n_{AM} - \mu_{AM} \quad (1)$$

where $E(n_{AM})$ is the total energy of the Gr/H-MoS_2 system with n_{AM} AM intercalated atoms, $E(0)$ is the energy of the pristine Gr/MoS_2 structure, and μ_{AM} is the chemical potential of the AM atom in the bulk structure (hcp for Li, Na, and K).

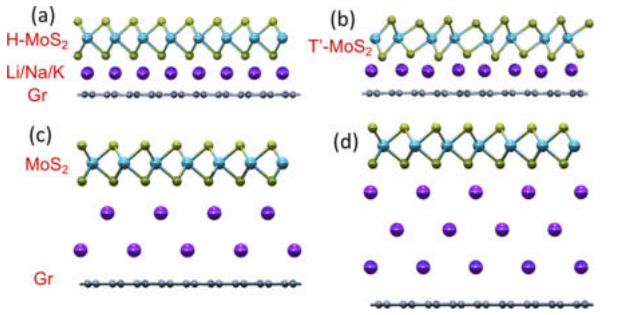


Figure 1. Typical atomic configurations illustrating the single- and multi-layer structures of AMs between the sheets of the host material, graphene (Gr) and MoS_2. (a,b) Single layer of AMs in the Gr/MoS_2 heterostructure with MoS_2 being in the H or T′ phase, respectively. hcp bilayer AM (c) and fcc trilayer (d) configurations in the Gr/MoS_2 heterostructure.

We also considered the reaction energy, E_r, the average gain or loss for adding one AM atom from the system. It is defined as

$$E_r = E(n_{AM} - 1) + \mu_{AM} - E(n_{AM}) \quad (2)$$

This is essentially the derivative of E_i with regard to n_{AM}. A positive value indicates that it is energetically favorable for additional AM atoms to be close to the existing AM structure. Negative values suggest that the intercalated AM atoms tend to be spatially separated.

RESULTS AND DISCUSSION

Prior to assessing the intercalation energy, we carried out the calculations to study the dependence of the results on the choice of the method to describe the vdW interaction. The proper account for vdW interactions is highly important when simulating multi-layer 2D materials and heterostructures,[34−36] with and without intercalated species. Specifically, the interlayer separation in the pristine system may change,[35] thus offering more or less space available for the intercalation. Moreover, the electronic properties, for example, the gap in the electronic spectrum may also substantially change, as demonstrated, for example, for platinum dichalcogenides.[37] The methods for the calculation of the vdW interaction can be divided into three categories: semiempirical models suggested by Grimme et al.,[38−40] models based on the atomic polarization modified by the electron density,[41−44] and full density functional approximations based on pairwise dispersion models using only the density.[45,46] The analysis of the methods for the prediction of the geometric and energy properties of various 2D materials (including graphene and MoS_2) was done by Tawfik et al.[36] Four of the eleven methods considered (MBD@rsSCS/FI + ER,[44] DFT-D3,[39] optB88-vdW,[47] and DFT-dDsC[43]) provide higher accuracy as compared to other methods. Also, the work of Tawfik et al.[36] reported that the most accurate method for a description of chemical and dispersion forces in 2D materials is SCAN + rVV10.[48] However, a critical issue with SCAN + rVV10 is its computational performance. We used several methods for calculating the vdW forces. The results of our calculations (μ_{Li}, E_i, and the average charge transfer from Li atoms to the host material) for 16 Li atoms in the system are listed in Table 1. All methods gave almost the same charge transfer (0.83−0.85 e^-), but the calculated E_i was quite different for different methods. For example, in the case of DFT-TS and DFT-TS/

https://doi.org/10.1021/acs.jpcc.2c03749
J. Phys. Chem. C 2022, 126, 15558−15564

Table 1. Computational Methods Used, Chemical Potential of Li Atom in the Bulk Structure μ_{Li}, Intercalation Energy E_i, and Charge Transfer Using Bader Analysis[50] in a Gr/MoS_2 Bilayer Heterostructure with 16 Li Atoms

method	μ_{Li}, eV/at.	E_i, eV/at.	e^- from Li
DFT-D2[38]	−2.02	−0.47	0.84
DFT-D3[39]	−2.00	−0.16	0.84
DFT-D3-BJ[40]	−2.07	−0.18	0.83
DFT-TS[42]	−4.14	1.01	0.83
DFT-TS/HI[51]	−4.02	1.91	0.84
DFT-dDsC[43]	−1.97	−0.08	0.84
optB86b-vdW[47]	−1.54	−0.17	0.84
optB88-vdW[47]	−1.51	−0.21	0.84
optPBE-vdW[47]	−1.49	−0.12	0.84
rev-vdW-DF2[52]	−1.60	−0.19	0.84
SCAN-rVV10[48]	−2.28	−0.27	0.85
vdW-DF2[53]	−1.41	−0.32	0.83
vdW-DF[45]	−1.48	0.12	0.84

HI methods, the formation energy is positive because this method yields μ_{Li} (for the bulk hcp structure taken as the reference) approximately 2 times lower than other methods, which was also shown previously.[49] Hence, taking into account the good prediction of energy (as compared to other more accurate methods) and lattice parameter of 2D materials, as well as computational performance, we used the zero damping DFT-D3 method of Grimme[39] in our calculations.

Having tested the vdW methods in the context of intercalation simulations, we first studied what positions of AMs are energetically more favorable and assessed the energetics of AM atoms between Gr/MoS_2 sheets in the single-layer configuration. It was previously shown[27] that in bilayer graphene with different stacking (AA stacking and AB stacking) the formation of a single-layer system with a geometry close to the C_6LiC_6 configuration with Li atoms being on top of the hexagons is possible. Here, we found that Li atoms take positions on top of the Mo atoms, in agreement with the results of a previous study.[18]

To find out the relationship between the energetics and concentration of AM atoms, and determine the most energetically favorable concentrations of Li, Na, and K atoms in the Gr/MoS_2 heterostructure, we calculated the total energy of the system with an increasing number n of AM atoms and computed E_i and E_r as functions of the AM atom concentration $x = n_{AM}/n_{Mo}$ where n_{Mo} is the total number of Mo atoms in the supercell (n_{Mo} = 16 in our calculations). Figure 2 shows the changes in E_i and E_r with increasing x. We would like to reiterate here that positive values of E_i indicate that intercalation is energetically unfavorable (although still possible if a voltage is applied to the system), and negative values of E_r indicate that segregation of the intercalated atoms into areas with different concentrations should be expected. In the case of Li, the maximum possible concentration of AM atoms in Gr/MoS_2 heterostructure is x = 1.00 (16 Li atoms per 16 Mo atoms), as $E_r < 0$ for higher concentrations, although E_i remains negative even at the highest concentration studied, Figure 2a. This result is in good agreement with the previously reported data.[18] The Gr/MoS_2 heterostructure can accommodate twice the number of Li atoms as compared to bilayer graphene.[54] As discussed previously,[54] the presence of the graphene layer makes the intercalated system more stable and allows cycling of Li without formation of lithium sulfide.

In the case of Na, the maximum possible concentration of AM atoms in Gr/MoS_2 heterostructure is x = 0.56, Figure 2b, and the maximum possible concentration of K is x = 0.5, Figure 2c. This can be associated with larger atomic radii of Na and K as compared to Li. We investigated the intercalation of Na and K in the Gr/MoS_2 heterostructure with x up to unity, but we found that for concentrations above $x \sim 0.81$ Na and K atoms form two-layer structures instead of the single-layer one and therefore the results for $x > 0.81$ are not included in the plots.

As shown in previous theoretical[55−57] and experimental[4,58,59] studies, an increase of the Li concentration in pure MoS_2 causes a transition from the trigonal H phase to an octahedral T phase and then to a distorted T′ phase. The transition from the H phase to the distorted T′ phases is associated with a substantial energy barrier which is considerably reduced by increasing the concentration of Li atoms. It is important to note that the H-MoS_2 layer is a semiconductor, whereas T′-MoS_2 is a metallic system. To understand what would happen in the Gr/MoS_2 heterostructure, we studied the influence of the AM concentration on the energetics of the H and T′ phases in the Gr/MoS_2 structure. To do so, we also considered intercalated AMs in Gr/T′-MoS_2 and calculated the energy difference ΔE between the Gr/H-MoS_2 and Gr/T′-MoS_2 heterostructures for each concentration of AMs. Figure 3 shows the dependence of ΔE on the concentration of Li, Na, and K atoms in the Gr/MoS_2 heterostructure with the MoS_2 sheet being in the H or T′ phase. As evident from the figure, in the case of Li at concentration $x \sim 1$, the T′ phase becomes more energetically favorable than the H phase. Upon Na and K intercalation, the H phase always remains preferable.

This phase transition from the H to the T′ phase is associated with charge transfer from Li atoms to the MoS_2 layer. It was theoretically demonstrated[56] that in the freestanding MoS_2 monolayer the phase transition occurs at the Li concentration of about x = 0.4 (note that as adsorption of Li atoms on both sides of the MoS_2 layer was considered in

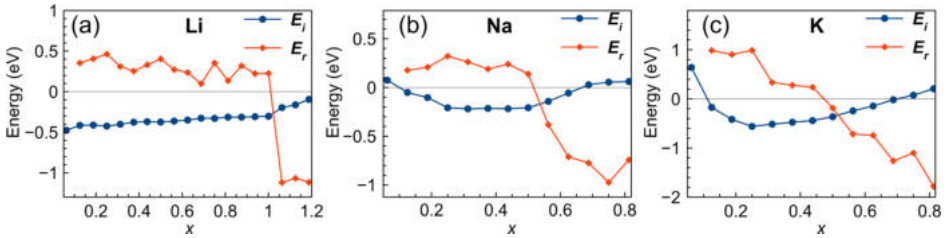


Figure 2. Intercalation energy E_i and reaction energy E_r as functions of the AM atom concentration $x = n_{AM}/N_{Mo}$ for Li (a), Na (b), and K (c) atoms intercalated as a single layer into a Gr/H-MoS_2 bilayer heterostructure.

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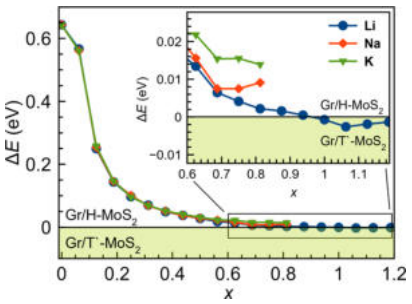


Figure 3. Difference in the total energy of Gr/H-MoS_2 and Gr/T′-MoS_2 heterostructures with intercalated AM atoms as a function of the AM concentration.

that paper, which increased the number of possible adsorption sites by a factor of 2, the concentration defined in our work should be divided by two for a comparison with the results presented in ref 56) and each Li atom donates approximately 0.8 e to the MoS_2 monolayer. Thus, 0.32 e per Mo atom should be sufficient for the transformation. In our simulations, only one side of MoS_2 is in contact with Li, and the charge which is donated by Li atoms is shared between the graphene and MoS_2 layers. We found that Li atoms intercalated between graphene and the MoS_2 layer donate approximately 0.8 e, Figure 4a,b, 0.5 e of which is accumulated on MoS_2 and 0.3 e

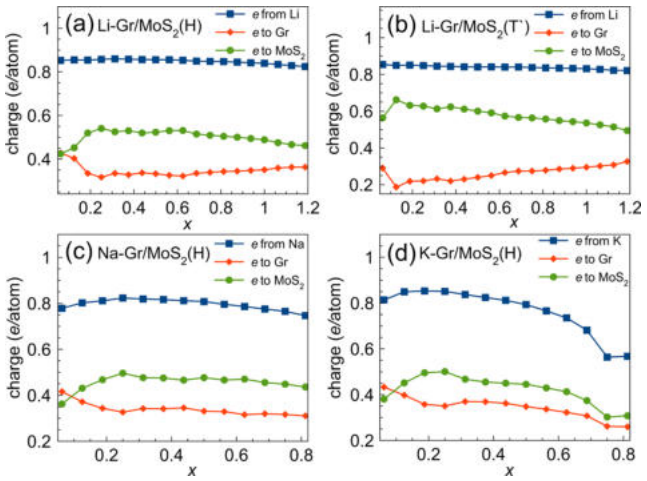


Figure 4. Charge transfer between the intercalated AM atoms and the host structure as a function of the concentration of AM atoms calculated using the Bader method for Li (a,b), Na (c), and K (d) for the Gr/H-MoS_2 and Gr/T′-MoS_2 heterostructures.

on graphene, in agreement with the previous studies.[17,18] The transition from the H to the T′ phase was found to occur at $x \sim 0.94$, which gives about 0.46 e per Mo atom, a value close but slightly higher than what was reported for the MoS_2 monolayer.

Using the Bader method, we extended the charge transfer analysis to K and Na. Figure 4c,d shows the dependence of the charge transfer from the AM atoms to graphene and MoS_2 on the AM atom concentration. In the case of Na, at a concentration ranging from 0.25 to 0.45 we observed the maximum number of electrons donated by Na (0.82 e) which matches the minimum of intercalation energy for this concentration, Figure 2b. Similar to Na, in the case of K the maximum number of electrons which leave K atoms (0.85 e, at $x = 0.25$) also matches the minimum of E_i, Figure 2c. In all systems (except for the case with one AM atom in the Gr/H-MoS_2 heterostructure) more electrons are accumulated in MoS_2.

To get a better understanding of the spatial redistribution of charge upon intercalation, we calculated the electron density difference $\Delta\rho$ (the density of the combined system minus the sum of the densities of the isolated heterostructure and single-layer AM atoms with the same atom positions as in the combined system) for single layers of AM atoms with different concentrations averaged in the planes parallel to the Gr/MoS_2 sheets as a function of the z-coordinate. The averaging was done by integrating the electron density within the plane parallel to the Gr/MoS_2 sheets. The results for Na are presented in Figure 5 for different concentrations correspond-

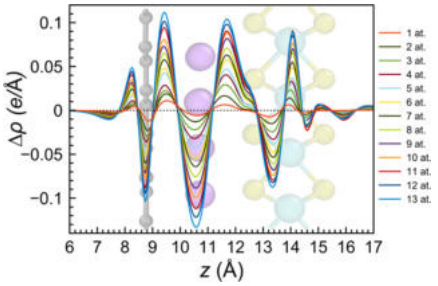


Figure 5. Electron density difference $\Delta\rho$ for a single layer of Na atoms with different concentrations (number of atoms) averaged in the planes parallel to the Gr/MoS_2 sheets as a function of the z-coordinate (perpendicular to the planes). The positions of the atoms are schematically shown.

ing to the number of AM atoms in the supercell ranging from 1 to 13. Areas below zero correspond to the depletion of electron density and above zero stand for density buildup. It is evident that the transferred charge is spatially localized mostly next to the S-layer of MoS_2 facing the AM atoms. According to these calculations, the charge accumulation, which can be associated with chemical bonding between Na atoms and MoS_2, is always larger than that with graphene except for the case of one AM atom between Gr/H-MoS_2. We received qualitatively similar results for other AMs that we considered (see the Supporting Information).

Because previous experimental[27] and theoretical[28] data indicated that formation of multi layers of AM atoms in bilayers of graphene and MoS_2 is possible, we also studied AM multi-layers in the Gr/H-MoS_2 heterostructure. To determine the most energetically favorable configurations of AM multi-layers, we considered different initial structures of the AM atoms. Initial atom positions were cut from the bulk fcc, hcp, and bcc metals along the low-energy directions (e.g., [111] for the fcc structures) and placed between the graphene and MoS_2 layers. The orientation of the metal crystal lattice with regard to graphene/MoS_2 and the sizes of the supercells were chosen to minimize the strain in the system. The strain normally did not exceed 1%. We calculated the intercalation energy and charge transfer for two, three, and four layers of Li, Na, and K in Gr/H-MoS_2 and Gr/T′-MoS_2 heterostructures. The intercalation energy as a function of the AM atom concentration for single- and multi-layer structures is shown in Figure 6a–c. In the case of Li, the intercalation energy is negative, which means that these structures can be formed. For Na, it is interesting that the intercalation energy for bcc and hcp multi-layers is negative and is roughly the same as for one

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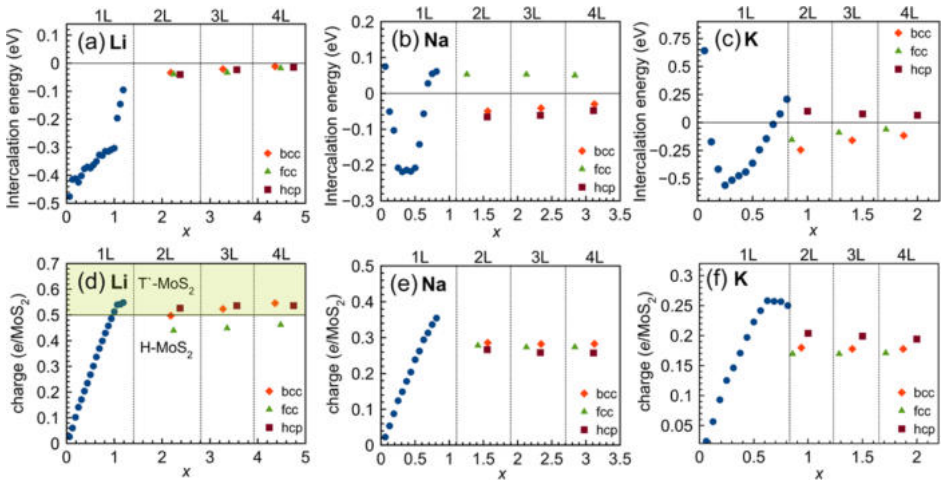


Figure 6. Intercalation energy as a function of the concentration of Li (a), Na (b), and K (c) atoms in single- (1L), double- (2L), and multi-layer (3L and 4L) configurations in the Gr/H-MoS_2 heterostructure. (d–f) Corresponding charge transfers to MoS_2 sheet in the heterostructure.

layer of Na with a concentration of atoms above 0.6. These results indicate that the formation of Na multi-layers in Gr/MoS_2 is possible for high concentrations of Na atoms. The same results were obtained for K, but for different structures—bcc and fcc.

As we showed previously for the intercalation of Li, see Figures 3 and 4a, the T′ phase becomes energetically favorable in MoS_2 at about 0.5 e per MoS_2 cell, so that it can also exist in multi-layer structures. As seen from Figure 6d, only in the case of Li, the multi-layers (in the case of bcc and hcp) donate the needed number of electrons to MoS_2. We compared the intercalation energy of all structures of Li multi-layers between Gr/H-Mos_2 and Gr/T′-MoS_2 and observed that in the case of bcc and fcc configurations the T′ phase is more energetically favorable. These results show that if multi-layer Li structures are formed, the phase transition from H to T′ in MoS_2 layer should indeed take place. Multi layers of Na and K do not donate enough electrons for the transition, Figure 6e,f, in agreement with our calculations of the total energies of the systems, which indicate that the H phase has lower energy.

CONCLUSIONS

To sum up, using DFT calculations, we systematically studied the intercalation of Li, Na, and K atoms into a graphene/MoS_2 heterostructure. We considered not only single-, but also multi-layer configurations of AM atoms, which have not been investigated before. We assessed the intercalation energetics for various concentrations of AM atoms, calculated charge transfer from AM atoms to the host system, and showed that although intercalation of AMs as a single layer is overall energetically preferable, multi-layer configurations can exist at high concentrations of AM atoms. We further demonstrated that the transition of the MoS_2 layer from the H to T′ phase is possible upon Li intercalation, but not for Na or K.

The simulations performed in this work were motivated by recent technical advances based on a peripheral galvanic cell geometry and Hall voltammetry that allow the study of intercalant motion, diffusion, and ordering without the complications brought about by the solid electrolyte interface in standard conventional battery cells.[27,60] The simulations offer important guidance as to which intercalants and heterointerfaces may be beneficial for further experimental explorations that aim at identifying systems with excellent mixed conductor properties as required for efficient anodes. Structural phase transformations may hamper reversibility and the awareness of their existence or absence is therefore valuable in the selection of promising intercalant/heterointerface combinations.

Overall, we expect that our results should help one to better understand the behavior of heterostructures assembled from 2D materials upon AM atom intercalation. We also envisage that our findings may stimulate further experimental effort focused on the tailoring of the electronic properties of such heterostructures. Moreover, the results of our simulations should be beneficial for increasing the capacity of anode materials in AM ion batteries.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcc.2c03749.

Parameters of the supercells used and the results of additional calculations (PDF)

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https://doi.org/10.1021/acs.jpcc.2c03749
J. Phys. Chem. C 2022, 126, 15558−15564

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank Dr. M. Ghorbani-Asl for fruitful discussion. A.V.K. acknowledges the German Research Foundation (DFG) for the support through project KR 4866/6-1 and the collaborative research center "Chemistry of Synthetic 2D Materials" SFB-1415-417590517. J.H.S. acknowledges financial support from the EU graphene flagship core 3 program. I.V.C. thanks the German Academic Exchange Service (DAAD) program "Mikhail Lomonosov" for support. The authors thank the HZDR Computing Center, HLRS, Stuttgart, Germany, and TU Dresden Cluster "Taurus" for generous grants of CPU time.

■ REFERENCES

(1) Xu, J.; Dou, Y.; Wei, Z.; Ma, J.; Deng, Y.; Li, Y.; Liu, H.; Dou, S. Recent Progress in Graphite Intercalation Compounds for Rechargeable Metal (Li, Na, K, Al)-Ion Batteries. *Adv. Sci.* **2017**, *4*, 1700146.

(2) Ji, K.; Han, J.; Hirata, A.; Fujita, T.; Shen, Y.; Ning, S.; Liu, P.; Kashani, H.; Tian, Y.; Ito, Y.; et al. Lithium intercalation into bilayer graphene. *Nat. Commun.* **2019**, *10*, 275.

(3) Sonia, F. J.; Jangid, M. K.; Ananthoju, B.; Aslam, M.; Johari, P.; Mukhopadhyay, A. Understanding the Li-storage in few layers graphene with respect to bulk graphite: experimental, analytical and computational study. *J. Mater. Chem. A* **2017**, *5*, 8662−8679.

(4) Bediako, D. K.; Rezaee, M.; Yoo, H.; Larson, D. T.; Zhao, S.; Taniguchi, T.; Watanabe, K.; Brower-Thomas, T. L.; Kaxiras, E.; Kim, P. Heterointerface effects in the electrointercalation of van der Waals heterostructures. *Nature* **2018**, *558*, 425−429.

(5) Ichinokura, S.; Sugawara, K.; Takayama, A.; Takahashi, T.; Hasegawa, S. Superconducting calcium-intercalated bilayer graphene. *ACS Nano* **2016**, *10*, 2761−2765.

(6) Durajski, A. P.; Skoczylas, K. M.; Szczęśniak, R. Superconductivity in bilayer graphene intercalated with alkali and alkaline earth metals. *Phys. Chem. Chem. Phys.* **2019**, *21*, 5925−5931.

(7) Liu, C.; Bai, Y.; Zhao, Y.; Yao, H.; Pang, H. MoS2/graphene composites: Fabrication and electrochemical energy storage. *Energy Storage Mater.* **2020**, *33*, 470−502.

(8) Shiva, K.; Ramakrishna Matte, H. S.; Rajendra, H. B.; Bhattacharyya, A. J.; Rao, C. N. Employing synergistic interactions between few-layer WS2 and reduced graphene oxide to improve lithium storage, cyclability and rate capability of Li-ion batteries. *Nano Energy* **2013**, *2*, 787−793.

(9) Thanh, T. D.; Chuong, N. D.; Hien, H.; Kshetri, T.; Kim, N. H.; Lee, J. H. Recent advances in two-dimensional transition metal dichalcogenides-graphene heterostructured materials for electrochemical applications. *Prog. Mater. Sci.* **2018**, *96*, 51−85.

(10) Huang, Z.; Han, X.; Cui, X.; He, C.; Zhang, J.; Wang, X.; Lin, Z.; Yang, Y. Vertically aligned VS 2 on graphene as a 3D heteroarchitectured anode material with capacitance-dominated lithium storage. *J. Mater. Chem. A* **2020**, *8*, 5882−5889.

(11) Chang, K.; Chen, W. In situ synthesis of MoS2/graphene nanosheet composites with extraordinarily high electrochemical performance for lithium ion batteries. *Chem. Commun.* **2011**, *47*, 4252−4254.

(12) Chang, K.; Chen, W. L-cysteine-assisted synthesis of layered MoS2/graphene composites with excellent electrochemical performances for lithium ion batteries. *ACS Nano* **2011**, *5*, 4720−4728.

(13) Chang, K.; Chen, W.; Ma, L.; Li, H.; Li, H.; Huang, F.; Xu, Z.; Zhang, Q.; Lee, J.-Y. Graphene-like MoS 2/amorphous carbon composites with high capacity and excellent stability as anode materials for lithium ion batteries. *J. Mater. Chem.* **2011**, *21*, 6251−6257.

(14) Pushparaj, R. I.; Cakir, D.; Zhang, X.; Xu, S.; Mann, M.; Hou, X. Coal-Derived Graphene/MoS2 Heterostructure Electrodes for Li-Ion Batteries: Experiment and Simulation Study. *ACS Appl. Mater. Interfaces* **2021**, *13*, 59950−59961.

(15) Miwa, R. H.; Scopel, W. L. Lithium incorporation at the MoS2/graphene interface: an ab initio investigation. *J. Phys.: Condens. Matter* **2013**, *25*, 445301.

(16) Ahmed, T.; Modine, N.; Zhu, J.-X. Bonding between graphene and MoS2 monolayers without and with Li intercalation. *Appl. Phys. Lett.* **2015**, *107*, 043903.

(17) Shao, X.; Wang, K.; Pang, R.; Shi, X. Lithium intercalation in graphene/MoS2 composites: first-principles insights. *J. Phys. Chem. C* **2015**, *119*, 25860−25867.

(18) Larson, D. T.; Fampiou, I.; Kim, G.; Kaxiras, E. Lithium intercalation in graphene−MoS2 heterostructures. *J. Phys. Chem. C* **2018**, *122*, 24535−24541.

(19) Sun, D.; Ye, D.; Liu, P.; Tang, Y.; Guo, J.; Wang, L.; Wang, H. MoS2/Graphene nanosheets from commercial bulky MoS2 and graphite as anode materials for high rate sodium-Ion batteries. *Adv. Energy Mater.* **2018**, *8*, 1702383.

(20) Li, G.; Luo, D.; Wang, X.; Seo, M. H.; Hemmati, S.; Yu, A.; Chen, Z. Enhanced Reversible Sodium-Ion Intercalation by Synergistic Coupling of Few-Layered MoS2 and S-Doped Graphene. *Adv. Funct. Mater.* **2017**, *27*, 1702562.

(21) David, L.; Bhandavat, R.; Singh, G. MoS2/graphene composite paper for sodium-ion battery electrodes. *ACS Nano* **2014**, *8*, 1759−1770.

(22) Xie, X.; Ao, Z.; Su, D.; Zhang, J.; Wang, G. MoS2/graphene composite anodes with enhanced performance for sodium-ion batteries: the role of the two-dimensional heterointerface. *Adv. Funct. Mater.* **2015**, *25*, 1393−1403.

(23) Massaro, A.; Pecoraro, A.; Muñoz-García, A. B.; Pavone, M. First-Principles Study of Na Intercalation and Diffusion Mechanisms at 2D MoS2/Graphene Interfaces. *J. Phys. Chem. C* **2021**, *125*, 2276−2286.

(24) Wang, B.; Peng, Y.; Yuan, F.; Liu, Q.; Sun, L.; Zhang, P.; Wang, Q.; Li, Z.; Wu, Y. A. A comprehensive review of carbons anode for potassium-ion battery: Fast kinetic, structure stability and electrochemical. *J. Power Sources* **2021**, *484*, 229244.

(25) Zhang, J.; Lai, L.; Wang, H.; Chen, M.; Shen, Z. X. Energy storage mechanisms of anode materials for potassium ion batteries. *Mater. Today Energy* **2021**, *21*, 100747.

(26) Zhang, P.; Yang, Y.; Duan, X.; Zhao, S.; Lu, C.; Shen, Y.; Shao, G.; Wang, S. Mechanistic investigations of N-doped graphene/2H(1T)-MoS2 for Li/K-ions batteries. *Nano Energy* **2020**, *78*, 105352.

(27) Kühne, M.; Börrnert, F.; Fecher, S.; Ghorbani-Asl, M.; Biskupek, J.; Samuelis, D.; Krasheninnikov, A. V.; Kaiser, U.; Smet, J. H. Reversible superdense ordering of lithium between two graphene sheets. *Nature* **2018**, *564*, 234−239.

(28) Chepkasov, I. V.; Ghorbani-Asl, M.; Popov, Z. I.; Smet, J. H.; Krasheninnikov, A. V. Alkali metals inside bi-layer graphene and MoS2: Insights from first-principles calculations. *Nano Energy* **2020**, *75*, 104927.

(29) Kresse, G.; Furthmüller, J. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B: Condens. Matter Mater. Phys.* **1996**, *54*, 11169.

(30) Kresse, G.; Joubert, D. From ultrasoft pseudopotentials to the projector augmented-wave method. *Phys. Rev. B: Condens. Matter Mater. Phys.* **1999**, *59*, 1758.

(31) Monkhorst, H. J.; Pack, J. D. Special points for Brillouin-zone integrations. *Phys. Rev. B: Solid State* **1976**, *13*, 5188.

(32) Momma, K.; Izumi, F. VESTA: a three-dimensional visualization system for electronic and structural analysis. *J. Appl. Crystallogr.* **2008**, *41*, 653−658.

(33) Smidstrup, S.; Markussen, T.; Vancraeyveld, P.; Wellendorff, J.; Schneider, J.; Gunst, T.; Verstichel, B.; Stradi, D.; Khomyakov, P. A.; Vej-Hansen, U. G.; et al. QuantumATK: An integrated platform of electronic and atomic-scale modelling tools. *J. Phys.: Condens. Matter* **2019**, *32*, 015901.

https://doi.org/10.1021/acs.jpcc.2c03749
J. Phys. Chem. C 2022, 126, 15558−15564

(34) Björkman, T.; Gulans, A.; Krasheninnikov, A. V.; Nieminen, R. M. van der Waals bonding in layered compounds from advanced density-functional first-principles calculations. *Phys. Rev. Lett.* **2012**, *108*, 235502.
(35) Björkman, T.; Gulans, A.; Krasheninnikov, A.; Nieminen, R. Are we van der Waals ready? *J. Phys.: Condens. Matter* **2012**, *24*, 424218.
(36) Tawfik, S. A.; Gould, T.; Stampfl, C.; Ford, M. J. Evaluation of van der Waals density functionals for layered materials. *Phys. Rev. Mater.* **2018**, *2*, 034005.
(37) Li, J.; Kolekar, S.; Ghorbani-Asl, M.; Lehnert, T.; Biskupek, J.; Kaiser, U.; Krasheninnikov, A. V.; Batzill, M. Layer-Dependent Band Gaps of Platinum Dichalcogenides. *ACS Nano* **2021**, *15*, 13249−13259.
(38) Grimme, S. Semiempirical GGA-type density functional constructed with a long-range dispersion correction. *J. Comput. Chem.* **2006**, *27*, 1787−1799.
(39) Grimme, S.; Antony, J.; Ehrlich, S.; Krieg, H. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu. *J. Chem. Phys.* **2010**, *132*, 154104.
(40) Grimme, S.; Ehrlich, S.; Goerigk, L. Effect of the damping function in dispersion corrected density functional theory. *J. Comput. Chem.* **2011**, *32*, 1456−1465.
(41) Tkatchenko, A.; DiStasio, R. A., Jr; Car, R.; Scheffler, M. Accurate and efficient method for many-body van der Waals interactions. *Phys. Rev. Lett.* **2012**, *108*, 236402.
(42) Tkatchenko, A.; Scheffler, M. Accurate molecular van der Waals interactions from ground-state electron density and free-atom reference data. *Phys. Rev. Lett.* **2009**, *102*, 073005.
(43) Steinmann, S. N.; Corminboeuf, C. A generalized-gradient approximation exchange hole model for dispersion coefficients. *J. Chem. Phys.* **2011**, *134*, 044117.
(44) Gould, T.; Lebègue, S.; Ángyán, J. G.; Bučko, T. A fractionally ionic approach to polarizability and van der Waals many-body dispersion calculations. *J. Chem. Theory Comput.* **2016**, *12*, 5920−5930.
(45) Dion, M.; Rydberg, H.; Schröder, E.; Langreth, D. C.; Lundqvist, B. I. Van der Waals density functional for general geometries. *Phys. Rev. Lett.* **2004**, *92*, 246401.
(46) Klimeš, J.; Bowler, D. R.; Michaelides, A. Van der Waals density functionals applied to solids. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2011**, *83*, 195131.
(47) Klimeš, J.; Bowler, D. R.; Michaelides, A. Chemical accuracy for the van der Waals density functional. *J. Phys.: Condens. Matter* **2010**, *22*, 022201.
(48) Peng, H.; Yang, Z.-H.; Perdew, J. P.; Sun, J. Versatile van der Waals density functional based on a meta-generalized gradient approximation. *Phys. Rev. X* **2016**, *6*, 041005.
(49) Park, J.; Yu, B. D.; Hong, S. Van der Waals density functional theory study for bulk solids with BCC, FCC, and diamond structures. *Curr. Appl. Phys.* **2015**, *15*, 885−891.
(50) Tang, W.; Sanville, E.; Henkelman, G. A grid-based Bader analysis algorithm without lattice bias. *J. Phys.: Condens. Matter* **2009**, *21*, 084204.
(51) Bučko, T.; Lebegue, S.; Hafner, J.; Angyan, J. G. Improved density dependent correction for the description of London dispersion forces. *J. Chem. Theory Comput.* **2013**, *9*, 4293−4299.
(52) Hamada, I. van der Waals density functional made accurate. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2014**, *89*, 121103.
(53) Lee, K.; Murray, É. D.; Kong, L.; Lundqvist, B. I.; Langreth, D. C. Higher-accuracy van der Waals density functional. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2010**, *82*, 081101.
(54) Shirodkar, S. N.; Kaxiras, E. Li intercalation at graphene/hexagonal boron nitride interfaces. *Phys. Rev. B* **2016**, *93*, 245438.
(55) Nasr Esfahani, D.; Leenaerts, O.; Sahin, H.; Partoens, B.; Peeters, F. Structural transitions in monolayer MoS2 by lithium adsorption. *J. Phys. Chem. C* **2015**, *119*, 10602−10609.
(56) Kan, M.; Wang, J.; Li, X.; Zhang, S.; Li, Y.; Kawazoe, Y.; Sun, Q.; Jena, P. Structures and phase transition of a MoS2 monolayer. *J. Phys. Chem. C* **2014**, *118*, 1515−1522.
(57) Kretschmer, S.; Komsa, H.-P.; Bøggild, P.; Krasheninnikov, A. V. Structural Transformations in Two-Dimensional Transition-Metal Dichalcogenide MoS2 under an Electron Beam: Insights from First-Principles Calculations. *J. Phys. Chem. Lett.* **2017**, *8*, 3061−3067.
(58) Kim, J. S.; Kim, J.; Zhao, J.; Kim, S.; Lee, J. H.; Jin, Y.; Choi, H.; Moon, B. H.; Bae, J. J.; Lee, Y. H.; et al. Electrical Transport Properties of Polymorphic MoS2. *ACS Nano* **2016**, *10*, 7500−7506.
(59) Kappera, R.; Voiry, D.; Yalcin, S. E.; Branch, B.; Gupta, G.; Mohite, A. D.; Chhowalla, M. Phase-engineered low-resistance contacts for ultrathin MoS2 transistors. *Nat. Mater.* **2014**, *13*, 1128−1134.
(60) Kühne, M.; Paolucci, F.; Popovic, J.; Ostrovsky, P. M.; Maier, J.; Smet, J. H. Ultrafast lithium diffusion in bilayer graphene. *Nat. Nanotechnol.* **2017**, *12*, 895−900.

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Supplementary Information to: Single- and Multi-Layers of Alkali Metal Atoms Inside Graphene/MoS_2 Heterostructures: a Systematic First-Principles Study

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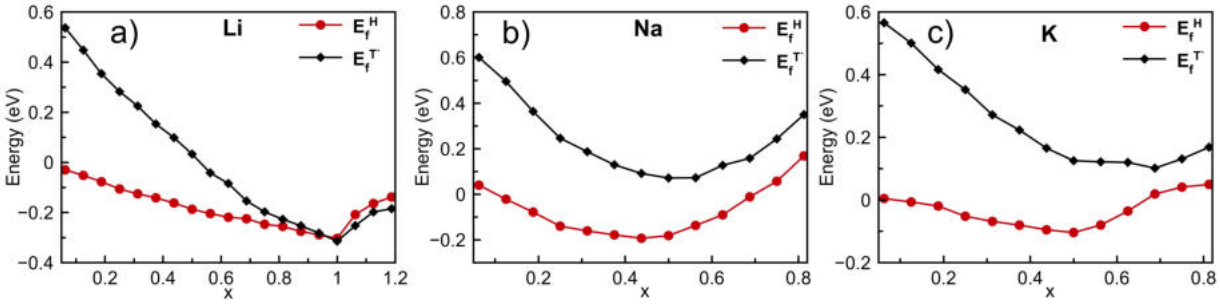


Figure S1: $E_f^{T'}$ and E_f^H as functions of the concentration of alkali metal atoms.

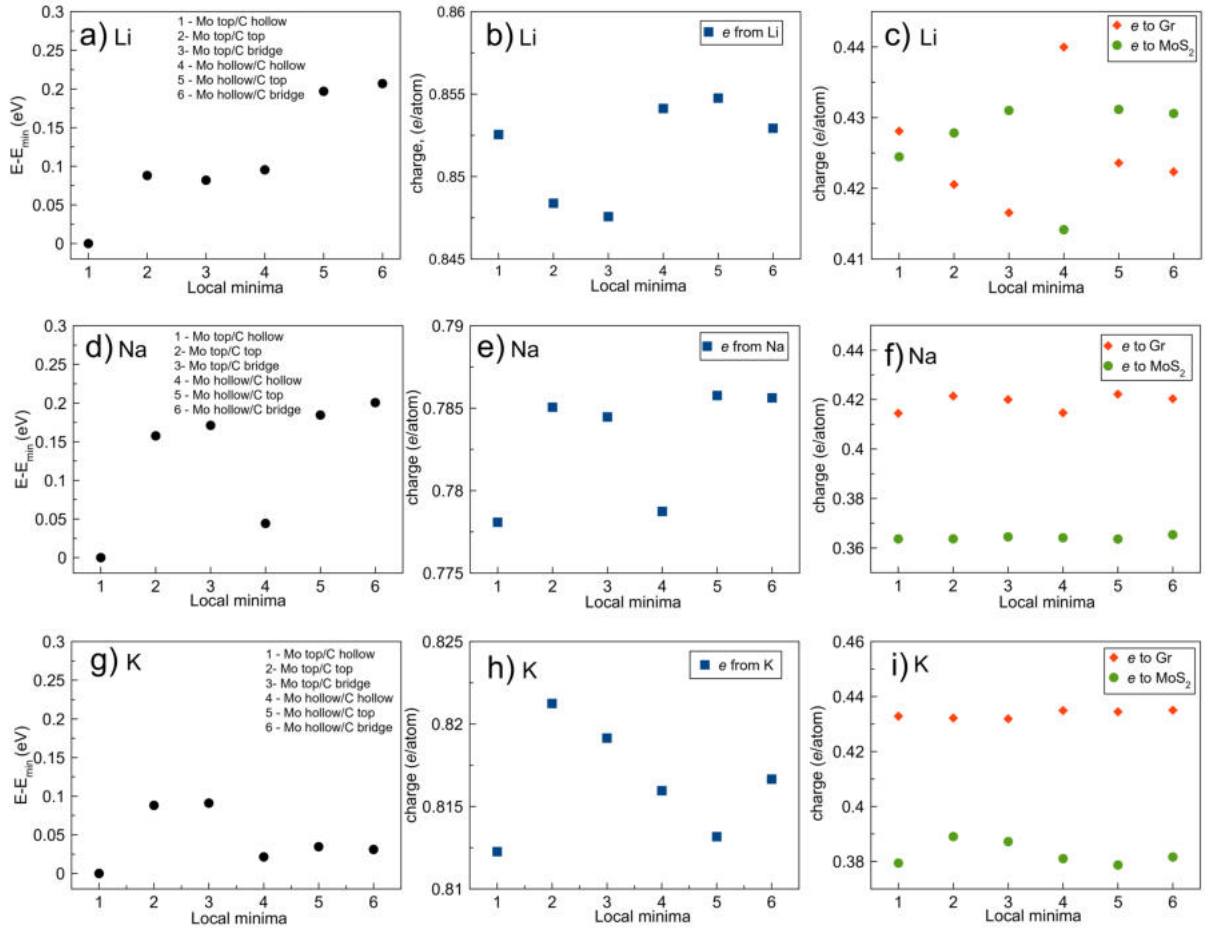


Figure S2: (a, d, g) Energy of a single Li, Na, K atoms in the bilayer Gr/H-MoS_2 structure with respect to the energy of the most stable configuration (location of the atom). (b, e, h) - charge donated by AM atoms, (c, f, i) - charge accumulated on MoS_2 and Gr.

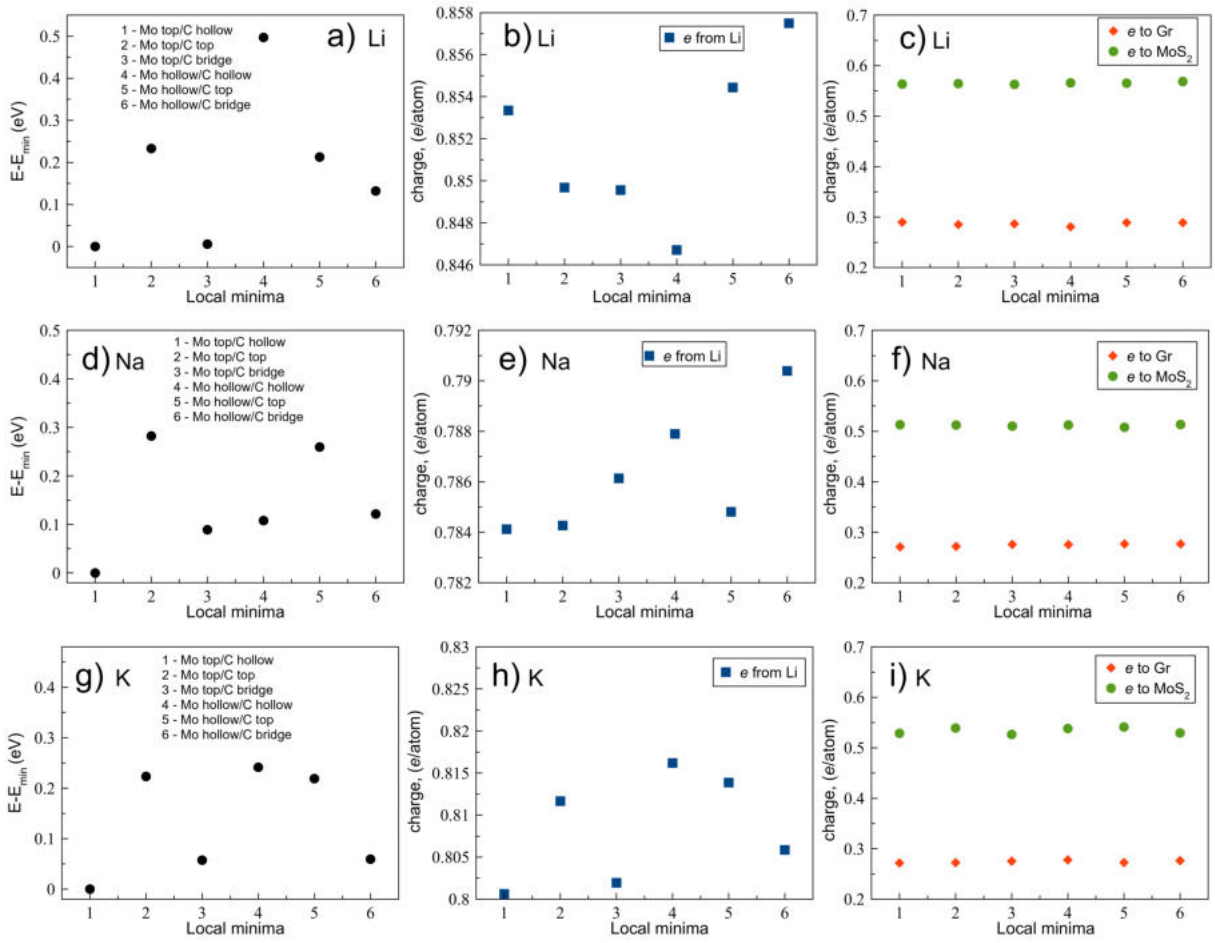


Figure S3: (a, d, g) Energy of a single Li, Na, K atoms in the bilayerGr/T′-MoS_2 structure with respect to the energy of the most stable configuration (location of the atom). (b, e, h) - charge donated by AM atoms, (c, f, i) - charge accumulated on MoS_2 and Gr.

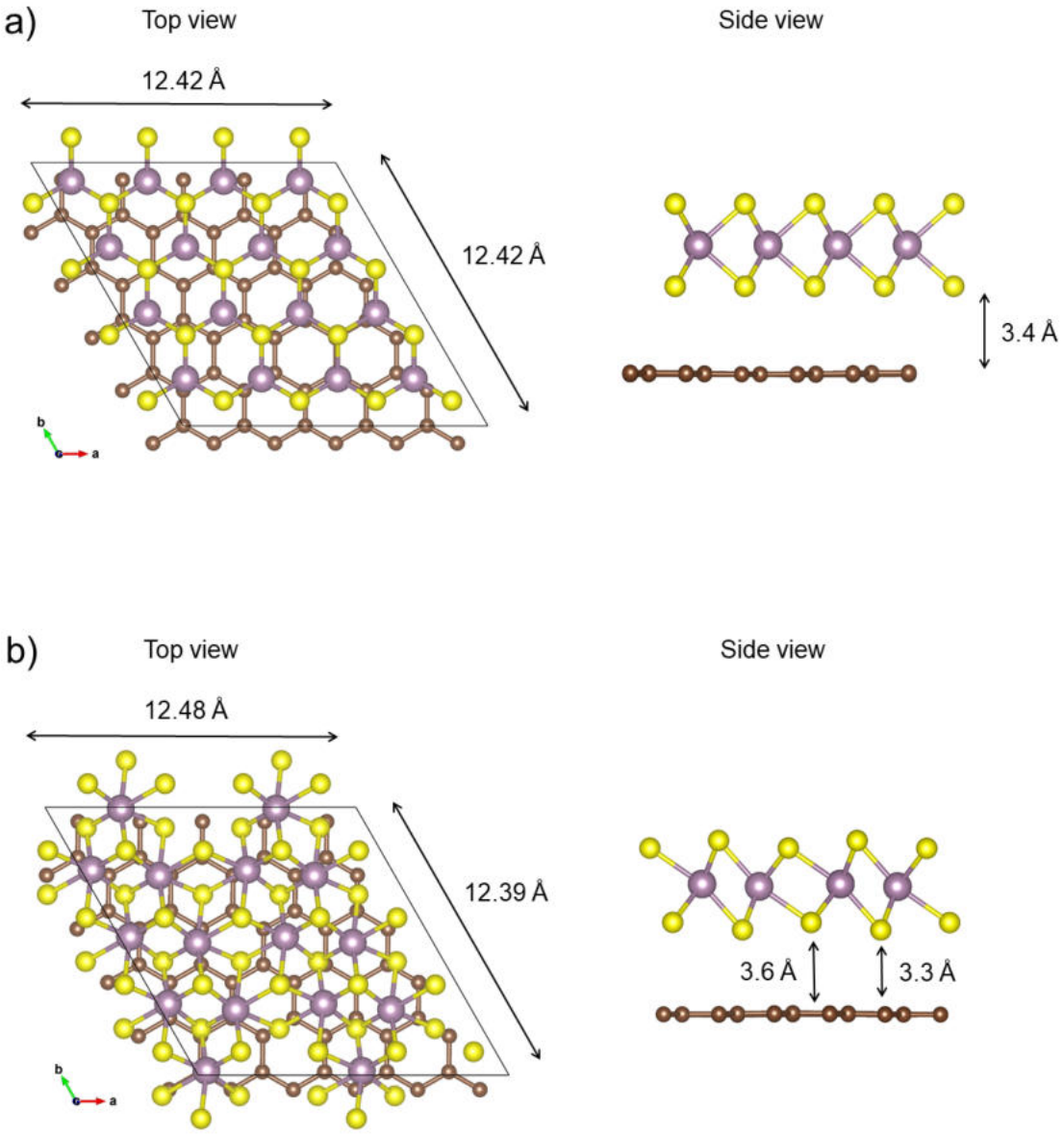


Figure S4: Top and side views of the Gr/H-MoS2 (a) and Gr/T′-MoS_2 (b) heterostructure.

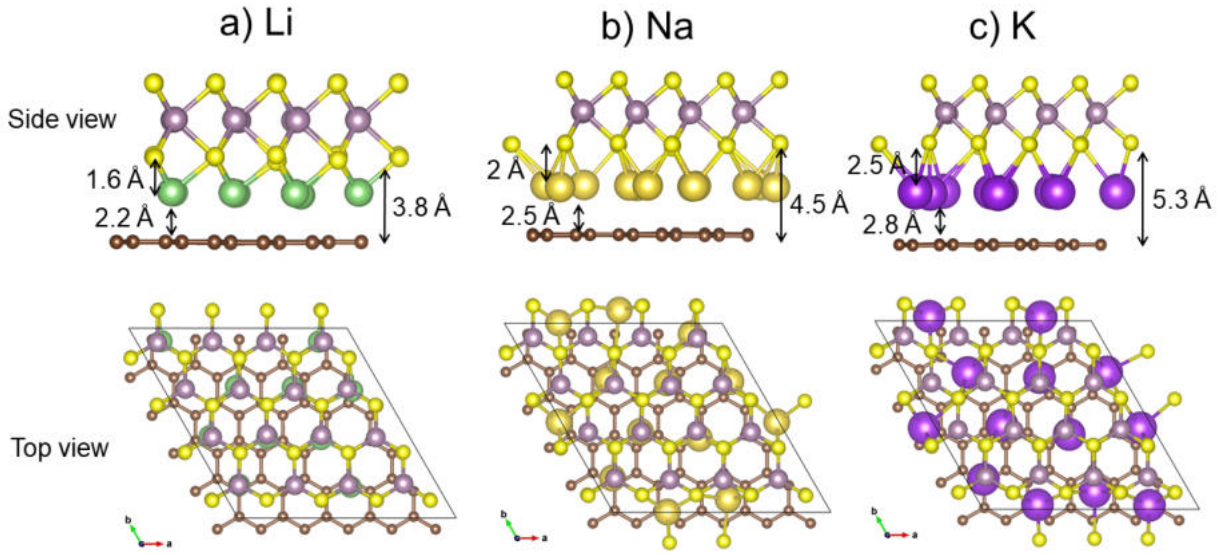


Figure S5: Top and side views Li (a), Na (b), K (c) atoms between the sheets of the host material, graphene (Gr) and MoS_2 with a concentration of x=0.625.

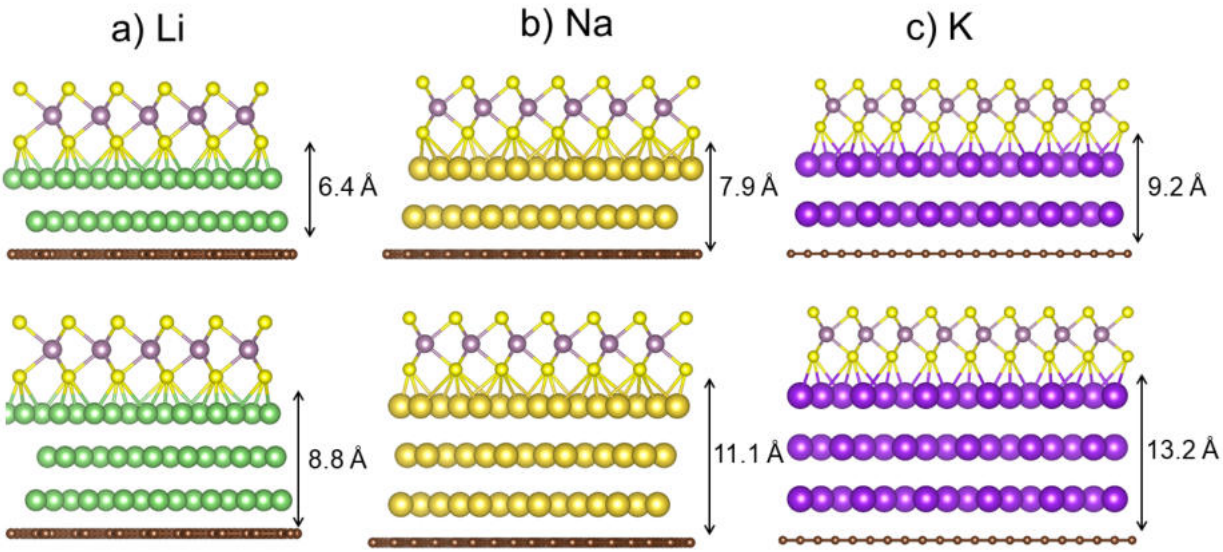


Figure S6: Side views of 2L and 3L of Li (a), Na (b), K (c) atoms between the sheets of the host material, graphene (Gr) and MoS_2

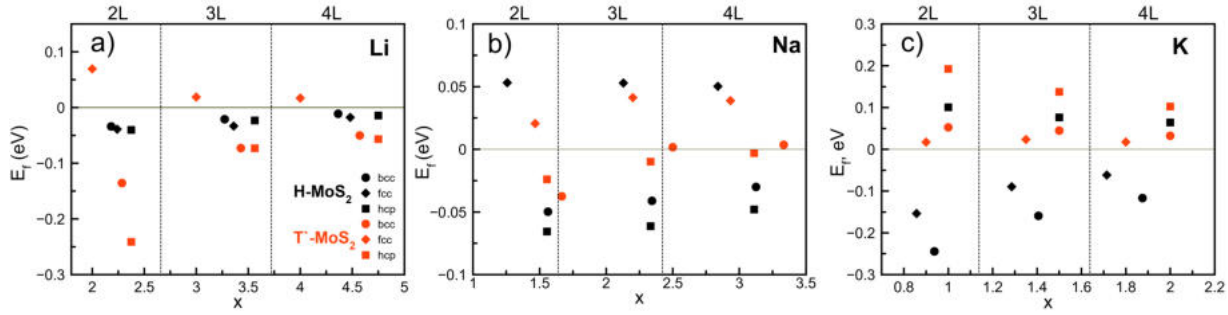


Figure S7: Formation energies of multy-layers Li, Na, K structures between Gr/H-MoS_2 and Gr/T′-MoS_2.

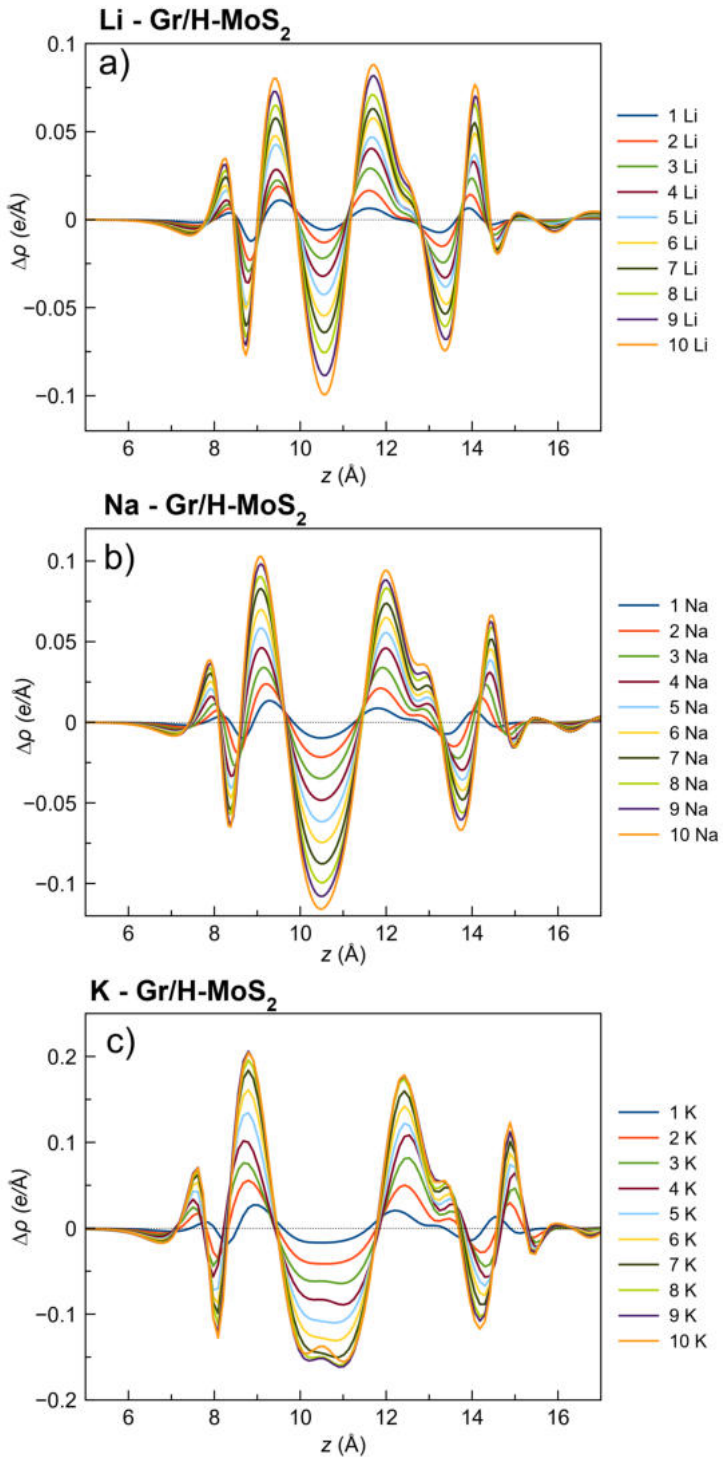


Figure S8: Electron density difference $\Delta\rho$ for a single layer of Li (a), Na(b) and K (C) atoms with different concentrations (number of atoms) averaged in the planes parallel to the Gr/MoS_2 sheets as a function of the z-coordinate (perpendicular to the planes).